



Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur

Choice Based Credit System (CBCS)
Structure and Syllabus
Multidisciplinary Minor Program in
Electronics and Computer Engineering
offered by
Department of Electronics Engineering
for Batch Admitted in

2024-25

Wamshi

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Walchand Institute of Technology
SOLAPUR-413006



Multidisciplinary Minor Program in Electronics and Computer Engineering

Legends Used

L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B.Tech.	Bachelor of Technology



Multidisciplinary Minor Program in Electronics and Computer Engineering

Course Code Format

2	4	E	C	U/P	2	C	C	1	T/L
Year of syllabus revision		Program Code		U-Under Graduate, P-Post Graduate	Semester No./ Year1/2/3/...8	Course Type		Course Serial No. 1-9	T-Theory, L-Lab session A-Tutorial P-Programming / Design

Program Code

ET	Electronics and Telecommunication Engineering
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Course Type

BS	Basic Science
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ES	Engineering Science
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HU	Humanities & Social Science
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MC	Mandatory Course
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CC	Programme Core Compulsory Course
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SN*	Self-Learning (<i>N* indicates the serial number of electives offered in the respective category</i>)
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EN*	Programme Core Elective Course (<i>N* indicates the serial number of electives offered in the respective category</i>)
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SK	Skill-Based Course
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SM	Seminar
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MP	Mini project
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PR	Project
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IN	Internship
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ON*	Open Elective (<i>N* indicates the serial number of electives offered in the respective category</i>)
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MD	Multidisciplinary Minor
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EM	Entrepreneurship/Economics/Management Courses
FP	Community Engagement Project / Field Project
AE	Ability Enhancement Course
VE	Value Education Course
IK	Indian Knowledge System
VS	Vocational & Skill Enhancement Course
RM	Research Methodology
HN	Honors' Degree Course
HR	Honors Research

Sample Course Code	
24ECU3MD6T	Analog Electronics



Multidisciplinary Minor Program in Electronics and Computer Engineering

Semester	Course Code	Name of Course	Engagement Hours			Credits					Total
			L	T	P		Theory	OE/ POE	ISE	ICA	
III	24ECU3MD6T	Analog Electronics	2	-	-	2	60	-	40	-	100
IV	24ECU4MD6T	Digital Electronics	1	-	-	1		-	50	-	50
V	24ECU5MD6T	Microcontrollers and Peripherals	3	-	-	3	60	-	40	-	100
VI	24ECU6MD6T	Advanced Controllers and Interfacing	2	-	-	2	60	-	40	-	100
VII	24ECU7MD6T	Electronic System Design	3	-	-	3	60	-	40	-	100
VII	24ECU7MD6A	Electronic System Design	-	1	-	1	-	-	-	25	25
		Sub-Total	11	1	0	12	240		210	25	475
Laboratory Courses											
IV	24ECU4MD6L	Digital Electronics	-	-	2	1	-	-	-	25	25
VI	24ECU6MD6L	Advanced Controllers and Interfacing	-	-	2	1	-	-	-	25	25
		Sub-Total	0	0	4	2	0		0	50	50
		Grand-Total	11	1	4	14	240		210	75	525



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Second Year B.Tech. (Multidisciplinary Minor in ECM Engineering, Semester-III)****24ECU3MD6T : Multidisciplinary Minor I –Analog Electronics****Teaching Scheme****Lectures** 2 Hours/week**Credits** 2**Examination Scheme****ESE** 60 Marks**ISE** 40 Marks**Introduction:**

This program basic course provides essential fundamental knowledge to the student to understand working of electronic devices, their analysis and designing of electronic applications using these electronic devices. It also covers design and analysis applications IC 555 timer and IC 741. This is one of the foundation courses which is vital for students to comprehend working of complex electronic circuits and systems.

Course Prerequisite:

The student needs to complete an introductory course in basic electronics which includes basic knowledge of diode, rectifiers, BJT and different BJT configurations. The student is expected to apply basic electrical theorems such as Kirchhoff's current law and Thevenin's theorem in electrical circuits to analyze the circuit performance.

Course Objectives:

1. To introduce concept of regulated and unregulated dc power supplies
2. To understand biasing techniques and applications of bipolar junction transistors.
3. To make aware Op-Amp characteristics, parameters, and various linear applications in analog signal processing
4. To understand the functioning of timer IC 555 and its applications.

Course Outcomes:

After completing this course, the student will be able to -

1. Differentiate between regulated and unregulated DC power supplies
2. Describe BJT biasing methods and their uses in circuits
3. Describe the characteristics and parameters of operational amplifiers and utilize them in linear analog signal processing circuits
4. Explain the working of timer IC 555 and make its use in various timing and pulse generation applications



Unit – I	Regulated power supply	05 Hours
Bridge rectifier, analysis of unregulated dc power supply using bridge rectifier and shunt capacitor filter. Regulated dc power supply, three pin regulators, construction of 5V and 12V regulated dc power supply using capacitor filter and three pin regulator IC 7805, 7812.		
Unit – II	Bipolar Junction Transistors (BJT) Application	05 Hours
Comparison of BJT configurations, BJT application as an amplifier - Voltage divider biasing, DC load line, operating point, thermal runaway, single stage CE amplifier, voltage gain and frequency response, BJT application as switch.		
Unit – III	Op-Amp and its applications	05 Hours
Op-Amp block diagram, and characteristics of Op-Amp, equivalent circuit of Op-Amp, practical Op-Amp, comparison of ideal and practical Op-Amp, Op-Amp parameters- input offset voltage, output offset voltage, input bias current, common mode rejection ratio, slew rate, power supply rejection ratio (PSRR). Op-Amp applications- Inverting, non-inverting amplifier, adder, subtractor, summing- scaling - averaging amplifier, integrator, differentiator, comparator.		
Unit – IV	IC555 Timer	05 Hours
IC 555- pin diagram, internal block diagram, Astable and Monostable multivibrator design using IC 555, design of timer application.		
Text Books		
1. Op-Amps and Linear Integrated Circuits, Ramakant A. Gayakwad, PHI Learning Pvt. Ltd., Fourth edition 2. Electronic Devices and Circuits, Allen Mottershed, PHI Publication 3. Electronic Devices and Circuits, David A.Bell, Oxford University, Press India, Fifth edition.		
Reference Books		
1. An introduction to Operational Amplifiers, Lucas M. Faulkenberry, John Wiley & Sons, Second edition. 2. Operational Amplifiers, G. B. Clayton, English Language Book Society, Second edition. 3. Electronic Devices and Circuits, Robert Boylestad, Prentice Hall International. 4. Electronic Circuit Design, S.N.Talbar, T.R.Sontakke, Sadhu Sudha Publications.		



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Second Year B.Tech. (Multidisciplinary Minor in ECM Engineering), Semester-IV****24ECU4MD6T : Multidisciplinary Minor II – Digital Electronics**

Teaching Scheme		Examination Scheme	
Lectures	1 Hours/week	ISE	50 Marks
Practical	2 Hours/week	ICA	25 Marks
Credits	2		

Introduction:

Digitization has spread to a wide range of applications, including information (computers), automation, communications, control systems and signal processing. This course on digital electronics provides a thorough understanding of digital system design. The course intends to cover combinational and sequential circuit analysis and design.

Course Prerequisite:

Student shall have knowledge of binary number system and logic gates. Student shall also have knowledge about basic electronics devices – diodes, BJT and FET.

Course Objectives:

1. To introduce to student concepts of digital logic circuits like number systems, Boolean algebra, operation of various gates etc. along with their applications.
2. To make student design combinational circuit.
3. To make student design sequential circuit.

Course Outcomes:

After completing this course, student will be able to –

1. Perform operations on BCD, gray codes and error- correcting codes (e.g. hamming code) for digital communication and error detection application.
2. Differentiate between logic families based on various performance characteristics
3. Simplify and implement Boolean functions using Boolean algebra, DeMorgan's laws, karnaugh maps and canonical forms (SOP, POS), while applying optimal gate-level design techniques using NAND & NOR gates.
4. Design and implement combinational circuits(e.g. adders, multiplexers, decoders) and sequential circuits(e.g. counters, registers, flip-flops) by utilizing appropriate design techniques

Unit – I	Number System and Codes	04 Hours
Review of number system, BCD codes, gray code, representation of signed numbers, error detecting and correcting codes – parity codes and hamming code, introduction to logic families, parameter definitions – noise margin, power dissipation, fan-in, fan-out, propagation delay, typical values for TTL,CMOS&ECL.		



Unit – II	Boolean Algebra and Logic Simplification	04 Hours
Theorems of Boolean algebra, DeMorgan's law, standard representation of logic functions – SOP, POS and canonical forms, simplification of logic functions – karnaugh maps, NAND and NOR implementations.		
Unit – III	Combinational Circuit Design	03 Hours
Design and analysis procedure, design of adder, subtractor and binary parallel adder, code conversion, design of multiplexers, demultiplexers, encoders, decoders and their applications, design of magnitude comparators and parity circuits.		
Unit – IV	Sequential Circuit Design	04 Hours
Latches - SR latch; Flip-flops: SR, JK, D, T and Master-slave, triggering of flip-flops, Flip-flop characteristic equations and excitation tables. Flip-flop applications –counters, registers		
Internal Continuous Assessment (ICA):		
- ICA shall consist of minimum 8 experiments based upon- - Verification of truth table of basic and universal gates. - Implementation of universal gates using basic gates. - Code conversion using logic gates: binary to gray, gray to binary, binary to excess-3. - Implementation of any one combinational circuit using multiplexer. - Implementation of any one combinational circuit using de-multiplexer. - Design a n-bit comparator using logic gates. - Perform 1's and 2's complement adder/subtraction using 4 bit parallel adder. - Convert J-K flip-flop to T (Toggle) flip-flop and D (Data) flip-flop. - Design and implement mod-n asynchronous counter. - Design and implement mod-n synchronous counter. - Design and implement a 4 bit bi-directional shift register.		
Text Books		
- Digital Fundamentals, Thomas L.Floyd, Global Edition-Pearson Education Limited - Fundamental of Digital Circuits-Anand Kumar-Prentice Hall of India Pvt.Ltd. - Digital Design, M.Morris Mano, PHI,Third edition - Modern Digital Electronics, R.P.Jain, TMH,Third edition		
Reference Books		
- Digital Design: Principles and Practices, Wakerly JohnF., Pearson Education, Forth edition - Digital Principles &Applications, Leach, Malvino, Sixth edition		



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Third Year B.Tech. (Multidisciplinary Minor in ECM Engineering), Semester-V****24ECU5MD6T : Multidisciplinary Minor III –
Microcontrollers and Peripherals****Teaching Scheme****Lectures** 3 Hours/week**Credits** 3**Examination Scheme****ESE** 60 Marks**ISE** 40 Marks**Introduction:**

This course introduces microcontrollers, their architecture, programming, and peripheral interfacing. It provides fundamental knowledge essential for embedded systems and real-world applications. The course is designed to be accessible to students by explaining concepts in a structured and simplified manner.

Course Prerequisite:

Students should have a basic understanding of digital electronics, logic gates, and basic C programming. Also Knowledge of Basics of Analog Electronics will help in better understanding of peripherals.

Course Objectives:

1. To introduce the basic concepts of microprocessors and microcontrollers.
2. To understand the internal architecture and working of the 8051 microcontroller.
3. To learn basic programming and interfacing techniques.
4. To explore communication protocols and real-world applications.

Course Outcomes:

After completing this course, student will be able to –

1. Differentiate between a microprocessor and a microcontroller.
2. Read and analyse the architecture and working of the 8051 microcontroller from datasheet
3. Write simple programs in Embedded C for microcontroller applications.
4. Interface LEDs, motors, and displays with a microcontroller.
5. Differentiate between communication protocols like RS232, I2C, and SPI.

Unit – I**Introduction to Microprocessors and
Microcontrollers****08Hours**

Basics of microprocessors: working principles, Address, data, and control buses and their role in data processing, Memory organization: RAM, ROM, Flash, EEPROM – types and classification, Differences between microprocessors and microcontrollers, Applications of microcontrollers in automation, robotics, and consumer electronics.



Unit – II	The 8051 Microcontroller Architecture	09 Hours
Overview of 8051 microcontroller: features and applications, Internal architecture: ALU, Boolean processor, oscillator, Registers: General-purpose and special function registers, Stack, stack pointer, and program counter, Clock and reset circuits and their importance, Memory organization: program memory vs. data memory, Pin configuration and I/O port operations, Addressing modes and instruction set overview.		
Unit – III	Microcontroller Programming	09 Hours
Basics of Embedded C programming and its advantages over assembly, GPIO (General-Purpose Input / Output) and port programming, Timers and counters: operation and applications, Interrupts and their role in multitasking, Serial communication basics and UART operation.		
Unit – IV	Peripherals	08 Hours
Interfacing LEDs and Seven-segment displays, LCD interfacing: character and graphical displays, Keypad and switch interfacing techniques, Basics of motor control: DC motor and Stepper motor interfacing, Practical applications of peripheral interfacing in embedded systems.		
Unit – V	Communication Protocols	08 Hours
Serial communication: RS232, I2C and SPI communication, Interfacing external memory with microcontrollers.		
Textbooks:		
1. Muhammad Ali Mazidi, “The 8051 Microcontroller and Embedded Systems,” Pearson. 2. Kenneth J. Ayala, “The 8051 Microcontroller,” Thomson.		
Reference books:		
1. Raj Kamal, “Embedded Systems: Architecture, Programming and Design,” McGraw Hill. 2. Frank Vahid, Tony Givargis, “Embedded System Design,” Wiley.		



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Third Year B. Tech. (Multidisciplinary Minor in ECM Engineering), Semester-VI****24ECU6MD6T Multidisciplinary Minor IV-
Advanced Controllers and Interfacing****Teaching Scheme****Lectures** **2 Hours/week****Practical** **2 Hours/Week****Credits** **3****Examination Scheme****ESE** **60 Marks****ISE** **40 Marks****ICA** **25 Marks****Introduction:**

This course introduces advanced microcontrollers like Arduino and ESP32, focusing on their architecture, programming, and real-world interfacing applications. It emphasizes hands-on learning, enabling students to develop embedded solutions using sensors, actuators, and communication protocol.

Course Prerequisite:

Students should have completed Microcontrollers and Peripherals and have basic knowledge of embedded programming.

Course Objectives:

1. To introduce the architecture and programming of Arduino and ESP32 microcontrollers.
2. To interface various sensors, actuators, and displays with microcontrollers.
3. To explore different communication protocols in systems.
4. To implement application based on sensor-actuator based applications using ESP32.

Course Outcomes:

After completing this course, student will be able to –

1. Differentiate between architectures and list features of Arduino and ESP32.
2. Write and debug embedded programs using IDE.
3. Interface sensors with microcontrollers.
4. Interface Actuators with microcontrollers.
5. Simulate Communication Protocols like RS232, I2C, SPI

Unit – I**Introduction to Advanced Microcontrollers****04 Hours**

Overview of Arduino and ESP32 microcontrollers, Comparison between Arduino (AVR) and ESP32 (Xtensa) architectures, Development environments: Arduino IDE, PlatformIO, Power management and low-power modes in ESP32.



Unit – II	Digital and Analog Interfacing	05 Hours
GPIO programming: Digital input and output, Pulse Width Modulation (PWM) and servo motor control, Analog input with ADC: Interfacing potentiometers and sensors.		
Unit – III	Sensor Interfacing	05 Hours
Interfacing temperature, humidity, and motion sensors, Ultrasonic sensor-based distance measurement. LDR Interfacing.		
Unit – IV	Actuator Interfacing	05 Hours
Stepper and DC motor control using motor drivers, Relay module interfacing for automation, LCD Display Interfacing.		
Unit – V	Communication Protocols	06 Hours
Serial communication: UART, Software Serial, I2C protocol: Interfacing multiple devices (OLED, EEPROM), SPI communication: Interfacing SD cards and displays, Bluetooth and BLE in ESP32.		
Internal Continuous Assessment (ICA):		
- ICA shall consist of minimum 8 experiments based upon- - Blinking LED and controlling GPIO using Arduino and ESP32 - Interface DHT Sensor to Read Temperature and Humidity. - Reading analog sensors like LDR and displaying data. - PWM control of LED brightness and servo motor positioning - Interfacing Ultrasonic Sensor to Measure Distance of Object. - Interfacing LCD/OLED display for sensor visualization - UART-based serial communication between Arduino/ESP32 and PC - I2C communication with peripherals (OLED, RTC module, etc) - Relay Module Interfacing to Control HV Appliances - DC Motor Interfacing with ESP32 using Driver Circuit		
Textbooks:		
- Programming Arduino: Getting Started with Sketches by Simon Monk, McGraw Hill. - Internet of Things with Arduino and ESP32 by Pradeeka Seneviratne, Packt.		
Reference Books:		
- Arduino Cookbook by Michael Margolis, O'Reilly Media. - Beginning Sensor Networks with Arduino and Raspberry Pi, Charles Bell,” Apress.		

